GREATEST COMMON FACTOR & LEAST COMMON MULTIPLE – LEVEL 1: EASY

Exercise 1

Find the Greatest Common Factor (GCF) and

Least Common Multiple (LCM) of:

$$24 \quad \text{and} \quad 36$$

Answer 1

1. Prime factorizations:

$$24 = 2^3 \cdot 3, \quad 36 = 2^2 \cdot 3^2$$

2. GCF (lowest powers): $2^2 \cdot 3 = 12$ 3. LCM (highest powers): $2^3 \cdot 3^2 = 72$

$$\texttt{GCF} = \mathbf{12}, \quad \texttt{LCM} = \mathbf{72}$$

Exercise 2

Find the Greatest Common Factor (GCF) of the

algebraic monomials:

$$18x^3y^2 \quad \text{and} \quad 30x^2y^4$$

Answer 2

1. Numerical GCF: GCF$(18, 30) = 6$ 2. Lowest power of x: x^2 3. Lowest power of y: y^2

$$\texttt{GCF} = \mathbf{6x^2y^2}$$

Exercise 3

Find the Least Common Multiple (LCM) of the

algebraic monomials:

$$12a^4b^2 \quad \text{and} \quad 15a^2b^5$$

Answer 3

1. Numerical LCM: LCM$(12, 15) = 60$ 2. Highest power of a: a^4 3. Highest power of b: b^5

$$\texttt{LCM} = \mathbf{60a^4b^5}$$

GREATEST COMMON FACTOR & LEAST COMMON MULTIPLE – LEVEL 2: MEDIUM

Exercise 4

Find the GCF and LCM of the three numbers:

$$45, \quad 60, \quad \text{and} \quad 75$$

Answer 4

1. Factorizations:

$$45 = 3^2 \cdot 5, \quad 60 = 2^2 \cdot 3 \cdot 5, \quad 75 = 3 \cdot 5^2$$

2. GCF $= 3 \cdot 5 = 15$ 3. LCM $= 2^2 \cdot 3^2 \cdot 5^2 = 900$

$$\texttt{GCF} = \mathbf{15}, \quad \texttt{LCM} = \mathbf{900}$$

Exercise 5

Find the LCM of the polynomial expressions:

$$6(x^2 - 9) \quad \text{and} \quad 9(x+3)^2$$

Answer 5

1. Factor fully:

$$6(x-3)(x+3) \quad \text{and} \quad 9(x+3)^2$$

2. Numerical LCM: LCM$(6, 9) = 18$ 3. Highest polynomial powers:

$$\texttt{LCM} = \mathbf{18(x-3)(x+3)^2}$$

Exercise 6

A baker has 48 chocolate cookies and 72 vanilla cookies. He wants to pack them into identical boxes with no cookies left over. What is the maximum number of boxes he can make?

Answer 6

1. Identify operation: Maximum boxes requires GCF.
2. Prime factors:

$$48 = 2^4 \cdot 3, \quad 72 = 2^3 \cdot 3^2$$

3. GCF$(48, 72) = 2^3 \cdot 3 = 24$

$$= \mathbf{24}\ \texttt{boxes}$$

Exercise 7

Three lighthouses flash their lights every 12, 18, and 30 seconds respectively. If they flash together at 8:00 AM, how many seconds later will they all flash together again?

Answer 7

1. Identify operation: Simultaneous repetition requires LCM. 2. Factorize times:

$$12 = 2^2 \cdot 3, \quad 18 = 2 \cdot 3^2, \quad 30 = 2 \cdot 3 \cdot 5$$

3. LCM$(12, 18, 30) = 2^2 \cdot 3^2 \cdot 5 = 180$

$$= \mathbf{180}\ \texttt{seconds (3 minutes)}$$

GREATEST COMMON FACTOR & LEAST COMMON MULTIPLE – LEVEL 3: HARD

Exercise 8

A rectangular floor measuring 420 cm by 336 cm is to be tiled completely with identical square tiles without cutting any. What is the largest possible side length of each square tile, and how many tiles will be needed?

Answer 8

1. Find GCF for tile size:

$$420 = 2^2 \cdot 3 \cdot 5 \cdot 7, \quad 336 = 2^4 \cdot 3 \cdot 7$$

$$\text{GCF}(420, 336) = 2^2 \cdot 3 \cdot 7 = 84 \text{ cm}$$

2. Calculate total tiles:

$$\text{Tiles} = \left(\frac{420}{84}\right) \times \left(\frac{336}{84}\right) = 5 \times 4 = 20$$

$$\boxed{\texttt{Side} = \mathbf{84}\ \texttt{cm}, \quad \texttt{Total} = \mathbf{20}\ \texttt{tiles}}$$

Exercise 9

Three runners start running around a circular track at 6:00 AM. Runner A completes a lap every 105 s, Runner B every 140 s, and Runner C every 175 s. At what exact time will all three meet again at the starting line?

Answer 9

1. Find LCM of times:

$$105 = 3 \cdot 5 \cdot 7, \quad 140 = 2^2 \cdot 5 \cdot 7, \quad 175 = 5^2 \cdot 7$$

$$\text{LCM} = 2^2 \cdot 3 \cdot 5^2 \cdot 7 = 2100 \text{ seconds}$$

2. Convert 2100 s to minutes: $2100/60 = 35$ min. 3. Add to start time: 6:00 AM + 35 min.

$$\boxed{= \mathbf{6:35}\ \texttt{AM}}$$

Exercise 10

A school humanitarian drive collected 210 apples, 252 juice boxes, and 294 granola bars. They are packed into identical kits. What is the maximum number of kits possible, and how many of each item will be in a single kit?

Answer 10

1. Find GCF of items:

$$210 = 2 \cdot 3 \cdot 5 \cdot 7, \; 252 = 2^2 \cdot 3^2 \cdot 7, \; 294 = 2 \cdot 3 \cdot 7^2$$

$$\text{GCF}(210, 252, 294) = 2 \cdot 3 \cdot 7 = 42 \text{ kits}$$

2. Items per kit:

$$\text{Apples} = 210/42 = 5, \quad \text{Juice} = 252/42 = 6$$

$$\text{Bars} = 294/42 = 7$$

$$\boxed{\mathbf{42}\ \texttt{kits}\ (\mathbf{5}\ \texttt{apples}, \mathbf{6}\ \texttt{juices}, \mathbf{7}\ \texttt{bars})}$$

Exercise 11

Two gears mesh together. Gear A has 42 teeth and Gear B has 60 teeth. Mark X is placed where two teeth meet. How many total teeth of Gear A must pass the contact point before mark X aligns again, and how many full rotations will Gear B make?

Answer 11

1. Find LCM of teeth:

$$42 = 2 \cdot 3 \cdot 7, \quad 60 = 2^2 \cdot 3 \cdot 5$$

$$\text{LCM}(42, 60) = 2^2 \cdot 3 \cdot 5 \cdot 7 = 420 \text{ teeth}$$

2. Rotations of Gear B:

$$\text{Rotations} = \frac{420}{60} = 7 \text{ full rotations}$$

$$\boxed{\mathbf{420}\ \texttt{teeth pass}, \quad \mathbf{7}\ \texttt{rotations of B}}$$